

DSRC Wave basic safety message relay study.

Reliability and latency effects Vs vehicular network density and mobility of safety messages simulated using ns-3 network simulator.

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Problem Statement:

In Vehicle Ad-hoc Networks (VANET) of high density, the massive relay of Basic Safety Messages (BSM) may saturate the transmission media in milliseconds due that it is not established a maximum Time-To-Live (TTL) for those broadcast messages. This causes packet loss and increment in latency, avoiding that vehicles beyond the safety distance to stop have enough time to react.

Research question:

How an scheme of TTL for BSM retransmission applied to multi-hop environment can optimize the number of un-necessary retransmissions without compromise the message delivery reliability?

Contribution:

This study develops a mechanism to implement TTL scheme that avoids infinite retransmission of BSM, avoiding channel saturation achieving message redundancy with a latency less than 100 ms in multi-hop VANETS up to 1 km.

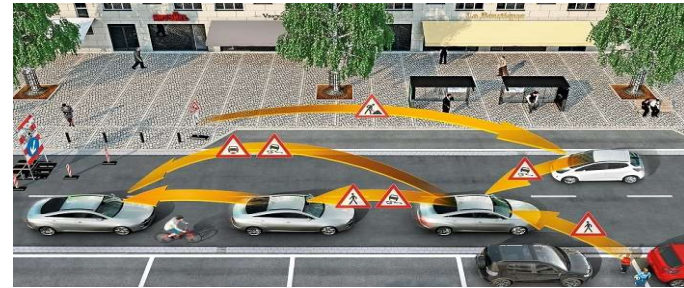


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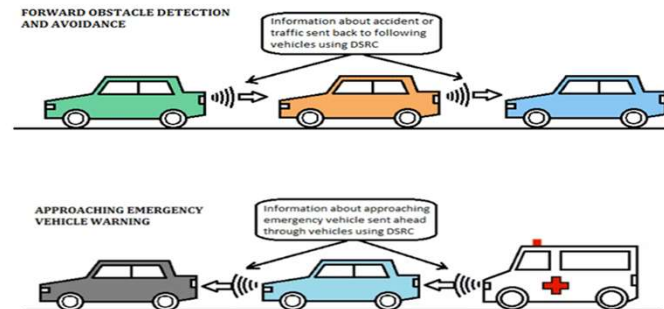
Wireless/mobile mechanisms:

Dedicated Short-Range Communications (DSRC, IEEE 802.11p) is a standard for vehicular networking (also known as WAVE for Wireless Access for Vehicular Environments), but real-world trials show that perfect packet reception is not the norm in VANETs (Vehicular Ad-hoc Networks):

- Losses occur due to distance, obstructions, and high mobility.
- In dense traffic or at high speeds, wireless contention and fading can lead to dropped safety messages, which is problematic for time-critical applications.
- It is prone to interferences (EMC/EMI).
- The message re-transmission due to packet loss can create delays in communication and throughput because there is not a defined Time-To-Live (TTL) for BSM, creating infinite retransmission looping



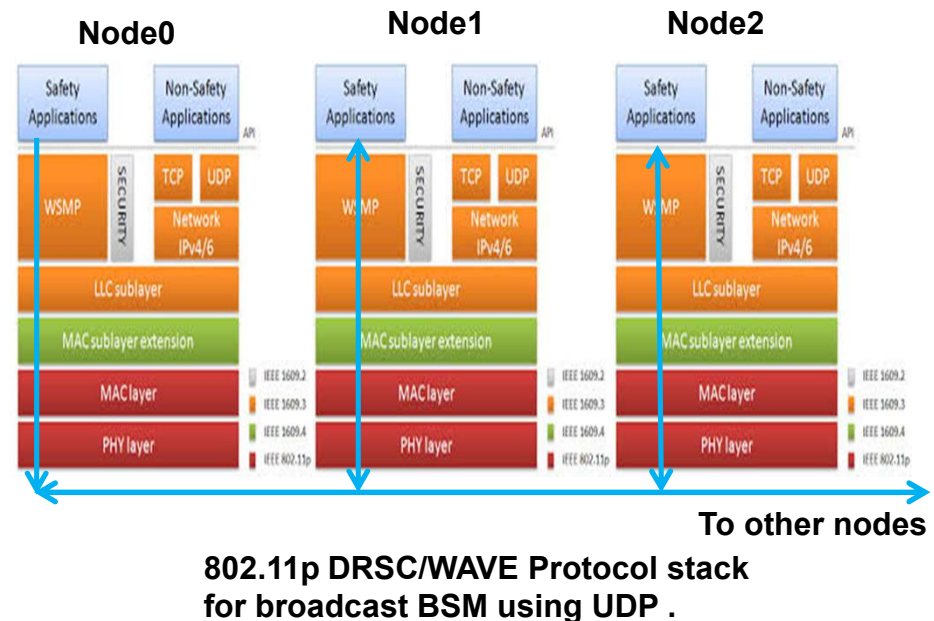
Dedicated Short Range Communication (DSRC).



Layered reasoning map:

The expected behavior is that by having a mechanism to discard a BSM after a number of retransmissions of those broadcast messages (“hops”) by all the vehicles on range, the transmission channel does not saturate.

- At the Application Layer (AP), the safety application identifies a high risk scenario that needs to be communicated using a BSM.
- The Wave Short Message Protocol (WSMP IEEE1609.3) integrates the TCP/IP layers for handle IP addressing and routing for efficient short message Tx and Rx.
- The Data Link Layer handles the frames into IEEE 802.11p with IEEE 1609.4 for manage the MAC and PHY layers.
- The other nodes share the same layers and will receive the packet, and if it is a BSM, it will be retransmitted also as a BSM. If this is continued forever, the transmission channel will be saturated. In order to avoid this, the TTL scheme needs to be implemented.



Hypotheses:

H_0 : Adding a TTL scheme to the IEEE 802.11p protocol, a BSM will reach at least 1 km away in a multi-hop high density scenario (urban environment of a highway) fast enough to allow the driver of the last car to react to stop the vehicle. To be tested by changing the density and the speed of vehicles with the TTL number.

H_1 : By increasing the transmitted power to the physical layer of the Wave IEEE 802.11p protocol, a BSM will reach at least 1 km away in a multi-hop high density scenario (urban environment of a highway) fast enough to allow the driver of the last car to react to stop the vehicle. To be tested by changing the Tx Power with different scenarios of the density and the speed of vehicles with the TTL number.



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Experimental design summary:

Using the ns3 network simulator version 3.37 with default parameters for WAVE libraries, it will be changed environmental and protocol parameters systematically:

- **Independent Variables:** Vehicle density (number of communicating cars within range), vehicle speed (e.g. city traffic vs. highway speeds), message load (10 Hz) and payload size of 200 bytes will not be changed to comply with WAVE BSM. It will may also be examined communication range by using different transmission power. Another variable is the use of a TTL for the relay scheme.
- **Dependent Metrics:** Packet delivery, end-to-end communication delay (time for a broadcast message to be received the last vehicle).
- **Baseline (Scenario 1):** A baseline scenario will be a best-case: low vehicle count and low speed, which should yield very high PDR (ideally ~100% for vehicles within close range) and minimal delay. This represents an ideal environment with minimal interference. It will be used this to benchmark performance against more challenging scenarios.



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Experimental design summary:

- **Evaluation Scenarios:** It will test at least four distinct scenarios:
 - **Scenario 1 (Baseline): Sparse Low-Speed Traffic** – Few vehicles (10) on a road, traveling at city speeds (40 km/h) separated by 100 m. This tests performance in light traffic with minimal contention and relative stability among nodes. It is expected high delivery ratios here (barring any distance issues).
 - **Scenario 2: Medium-Speed and Medium-Density Traffic** – Some vehicles (20) at medium speeds (80 km/h) separated by 50 m. This simulates a calm local highway. It will be observed increased contention and increased packet loss and latency
 - **Scenario 3: High-Speed and Medium-Density Traffic** – Some vehicles (20) at medium speeds (120 km/h) separated by 50 m. This simulates a normal highway. It will be observed increased contention and increased packet loss and latency
 - **Scenario 4: Low-Speed and High-Density Traffic** – Several vehicles (100) at low speeds (20 km/h) separated by 10 m. This simulates a congested highway. It will be observed increased contention and increased packet loss and latency.

Distance to stop Vs Range achieved by Tx Power:

- Based on commonly used criteria, the formula for estimate the total stopping distance is:

$$D_{total} = D_{p-r} + D_{braking} = vt_{p-r} + \frac{v^2}{2\mu g}$$

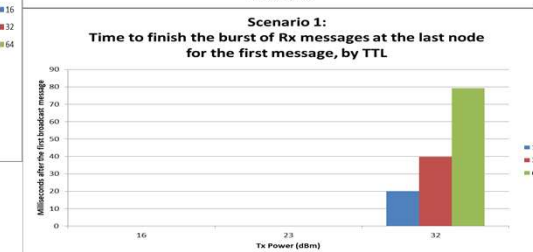
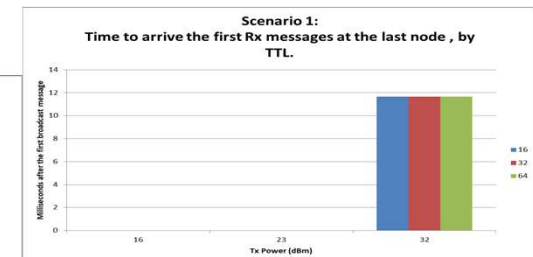
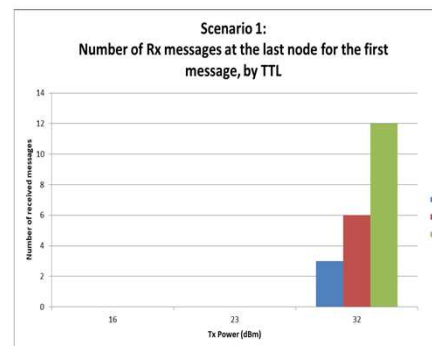
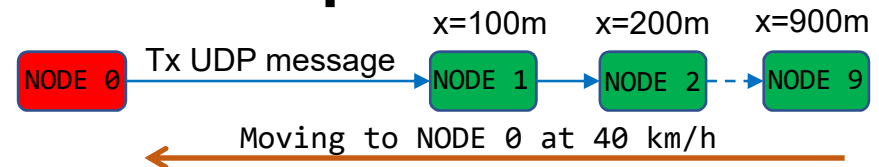
- Where:
 - The delay to response (D_{p-r}) is defined by the current vehicle speed (v) and the standard response driver time ($t_{p-r} = 1.5$ s).
 - The standard coefficient constant ($\mu = 0.7$).
 - Gravity ($g = 9.81$ m/s²).
- The results show that for different speeds, the distance and time to stop increases with speed:
 - For 88.61 km/h (55 mi/h) is 125.14 m (2.0 seconds to stop).
 - For 104.58 km/h (65 mi/h) is 166.46 m (2.7 seconds to stop).
 - For 120.67 km/h (75 mi/h) is 213.89 m (3.4 seconds to stop).
- Using the default Log Distance Propagation Loss model with exponent 3, and using the Tx Power levels of 16 dBm, 23 dBm and 32 dBm the range achieved using the default detection threshold of -62 dBm are 51.3 m, 87.9067 m and 175.4 m). This can be adjusted to a more realistic scenario if needed.



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Baseline Scenario 1: Sparse Low-Speed Traffic

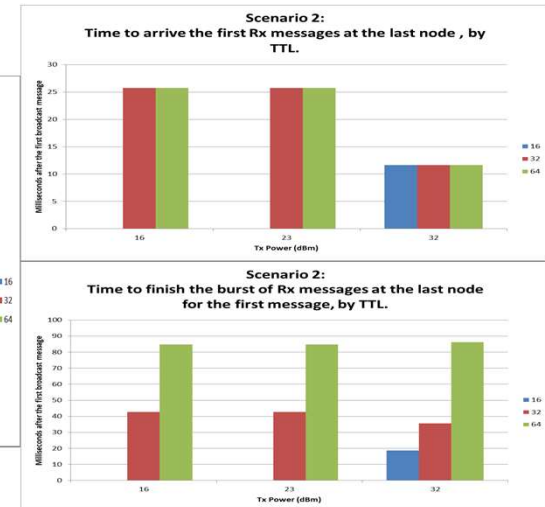
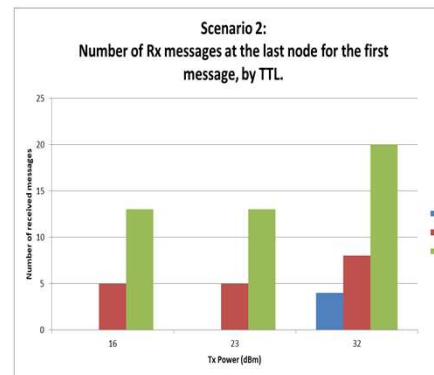
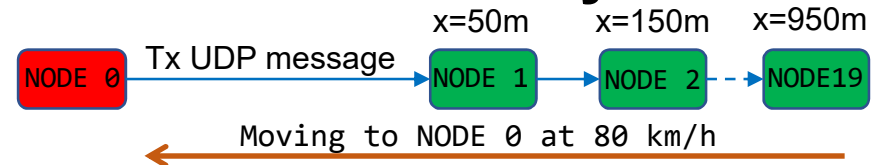
- The NODE 0 is fixed at the origin ($x = 0$ m), while the NODE 1 to NODE 9 will be at the start of the simulation with distances between them of 100 m, up to 900 m. Speed is 40 km/h.
- The ns3 propagation models are Log Distance Propagation Loss model and Constant Speed Propagation Delay. Packet detection threshold is -62 dBm by default. Doppler effect is ignored.
- NODE 0 is broadcasting BSM UDP packets at 6 Mbps of size=200 bytes, and transmission rate of 100 ms, with TxPower = 16, 23 and 32 dBm.
- NODEs 1 to 8 are rebroadcasting the BSM with delay of 1ms, while NODE 9 is used to log the time and distance when the BSM was received.
- The simulation results show that the total distance to the hazard is covered with TxP = 32 dBm because it is the only one that is able to reach the distance between the nodes.
- If TTL is higher, more messages reach the Node 9, having more dispersion close to the next message timeframe.



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Scenario 2: Medium-Speed and Medium-Density Traffic

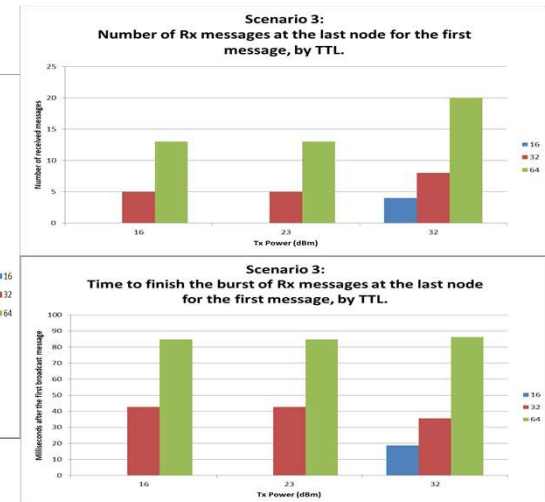
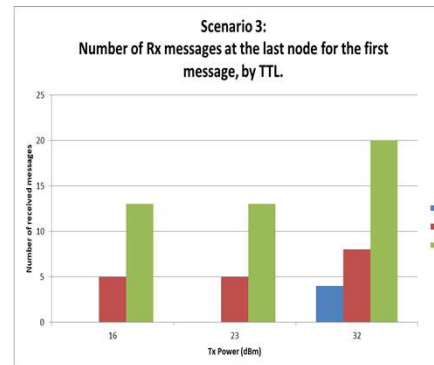
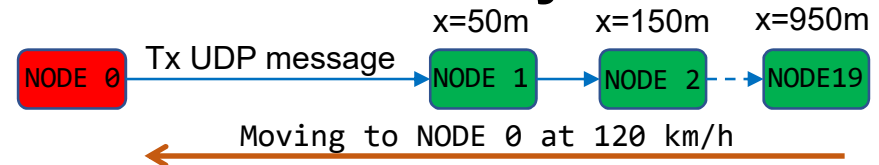
- The NODE 0 is fixed at the origin ($x = 0$ m), while the NODE 1 to NODE 19 will be at the start the simulation with distances between them of 50 m, up to 950 m. Speed is 80 km/h.
- The ns3 propagation models are Log Distance Propagation Loss model and Constant Speed Propagation Delay. Packet detection threshold is -62 dBm by default. Doppler effect is ignored.
- NODE 0 is broadcasting BSM UDP packets at 6 Mbps of size=200 bytes, and transmission rate of 100 ms, with TxPower = 16, 23 and 32 dBm.
- NODEs 1 to 8 are rebroadcasting the BSM with delay of 1ms, while NODE 19 is used to log the time and distance when the BSM was received.
- The simulation results show that the total distance to the hazard is covered with any TxP dBm because all options are able to reach beyond the distance between the nodes.
- If TTL is lower than the minimum hops to cover, the BSM messages does not reach the Node 19, having less delay with TTL = 64.



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Scenario 3: High-Speed and Medium-Density Traffic

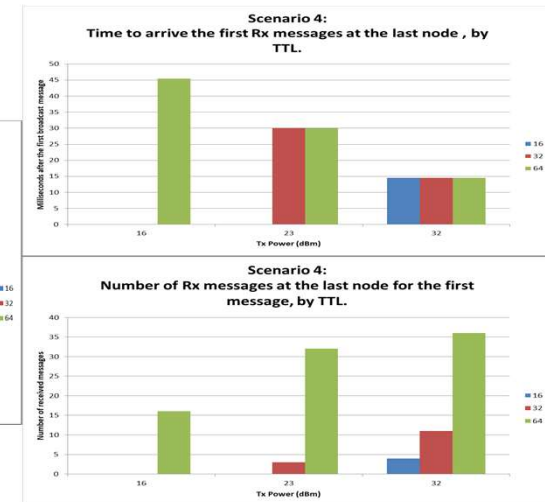
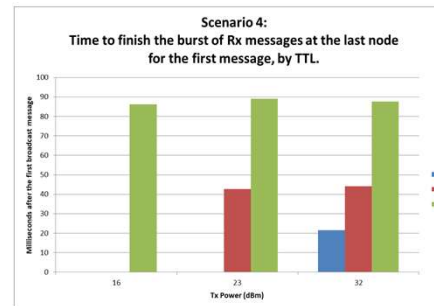
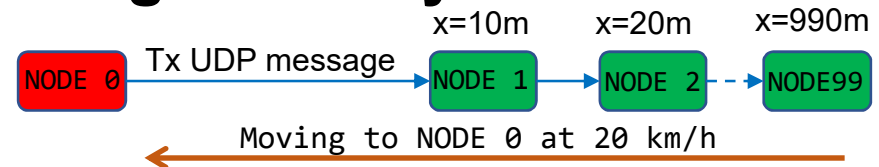
- The NODE 0 is fixed at the origin ($x = 0$ m), while the NODE 1 to NODE 19 will be at the start the simulation with distances between them of 50 m, up to 950 m.
- The ns3 propagation models are Log Distance Propagation Loss model and Constant Speed Propagation Delay. Packet detection threshold is -62 dBm by default. Doppler effect is ignored.
- NODE 0 is broadcasting BSM UDP packets at 6 Mbps of size=200 bytes, and transmission rate of 100 ms, with TxPower = 16, 23 and 32 dBm.
- NODEs 1 to 8 are rebroadcasting the BSM with delay of 1ms, while NODE 19 is used to log the time and distance when the BSM was received.
- The simulation results show that the total distance to the hazard is covered with any TxP dBm because all options are able to reach beyond the distance between the nodes.
- If TTL is lower than the minimum hops to cover, the BSM messages does not reach the Node 19, having less delay with TTL = 64.



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Scenario 4: Low-Speed and High-Density Traffic

- The NODE 0 is fixed at the origin ($x = 0$ m), while the NODE 1 to NODE 99 will be at the start the simulation with distances between them of 10 m, up to 990 m. Speed is 20 km/h.
- The ns3 propagation models are Log Distance Propagation Loss model and Constant Speed Propagation Delay. Packet detection threshold is -62 dBm by default. Doppler effect is ignored.
- NODE 0 is broadcasting BSM UDP packets at 6 Mbps of size=200 bytes, and transmission rate of 100 ms, with TxPower = 16, 23 and 32 dBm.
- NODEs 1 to 8 are rebroadcasting the BSM with delay of 1ms, while NODE 19 is used to log the time and distance when the BSM was received.
- The simulation results show that the total distance to the hazard is covered with any TxP dBm because all options are able to reach beyond the distance between the nodes.
- If TTL is lower than the minimum hops to cover, the BSM messages does not reach the Node 19, having less delay with TTL = 64.



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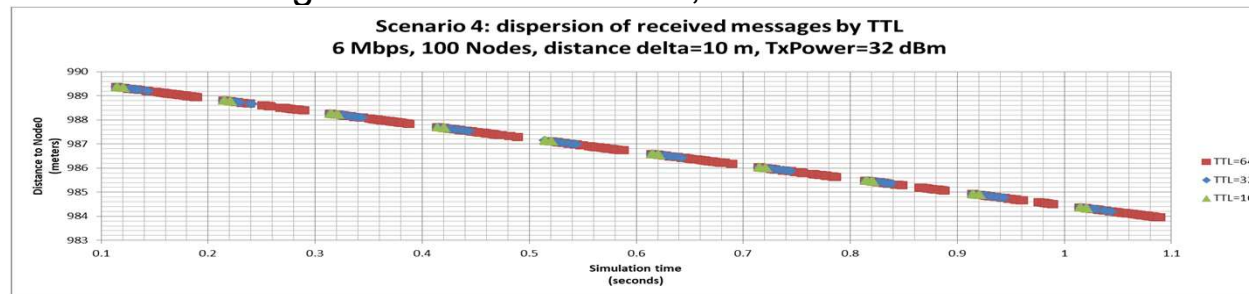
Limitations:

- The main limitation on this simulation is the default propagation model used (LogDistancePropagationLossModel), which is the one recommended for WAVE. It sets the detection threshold (CCaEdThreshold) at -62 dBm, which limits the distance range to achieve to 175.4 m with Tx Power of 32 dBm.
- In the literature it is documented that in real world scenarios is reaching more than 300 m with 20 dBm. That can be solved by artificially increasing the Tx Power or decreasing the detection threshold until more than 300 m is reached, but it is beyond the scope of this study that it is more focused into TTL effects and multi-hopping (with the direct effect that only 3 hops will be needed to reach 1 km).
- Another limitation is that ns3 models does not add the Doppler effect in the propagation model, that can be a big problem for high relative speeds (160-400 km/h) causing substantial shifts and broaden spectrum in OFDM signals causing Inter-Carrier Interference (ICI) and channel degradation.

Tradeoffs:

The implemented TTL scheme in ns3 is that each time that a re-broadcast of the BSM message will be done, the TTL tag number is decreased by one, and if it has reached 0, it is not retransmitted anymore.

- Advantage: If the TTL time is increased, more “hops” can be done, and the probability of reach further vehicles (nodes in ns3) is increased because it is needed at least the same number of TTL as hops to reach a node.
- Disadvantage: When a broadcast message is transmitted, not only the next nodes in range can receive it, also all nodes backwards the intended direction in range will also receive it, which also will broadcast again that message until TTL gets to zero. The effect of this is that more than one message will reach the end of the chain, with different delays creating a “burst” of received messages dispersed in time that it may overlap the next timeframe of 100 ms for the next broadcasted message that it may have different safety information (not an issue if the range is more than 300 m, where TTL=16 is a safe number).



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